LECTURE 1:

MODELS OF SYSTEMS AND THE MODELING PROCESS

OUTLINE

1.1 Systems, Models, and Modeling

1.2 Uses of Scientific Models

1.3 Example: Island Biogeography

1.4 Classification of Models

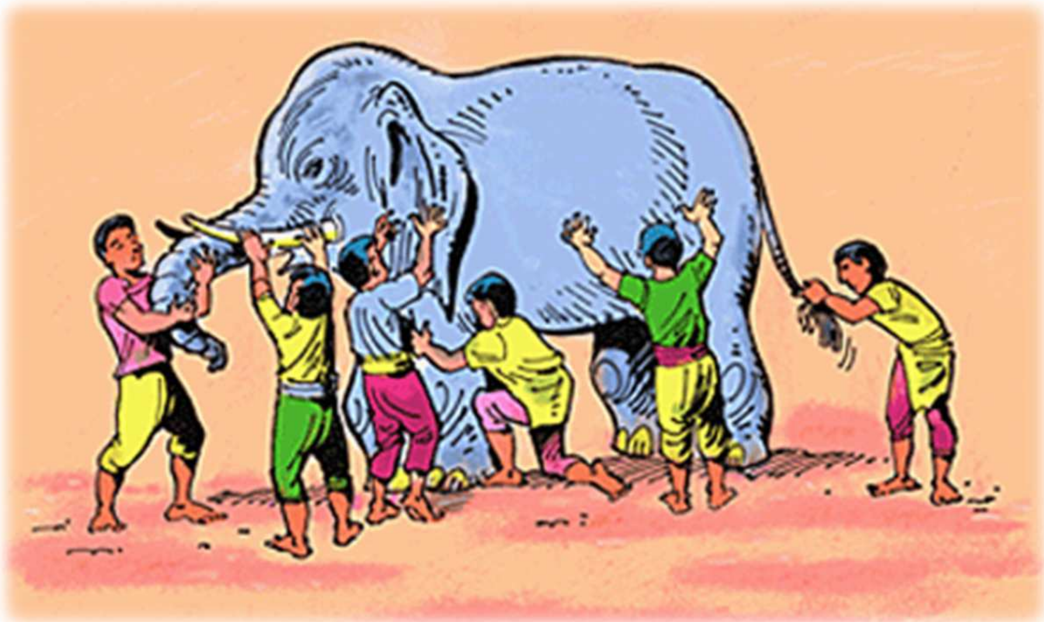
1.5 Constraints on Model Structure

1.6 Some Terminology

1.7 Misuses of Models: The Dark Side

1.1 Systems, Models, and Modeling

- The famous parable of six blind man inspecting an elephant.

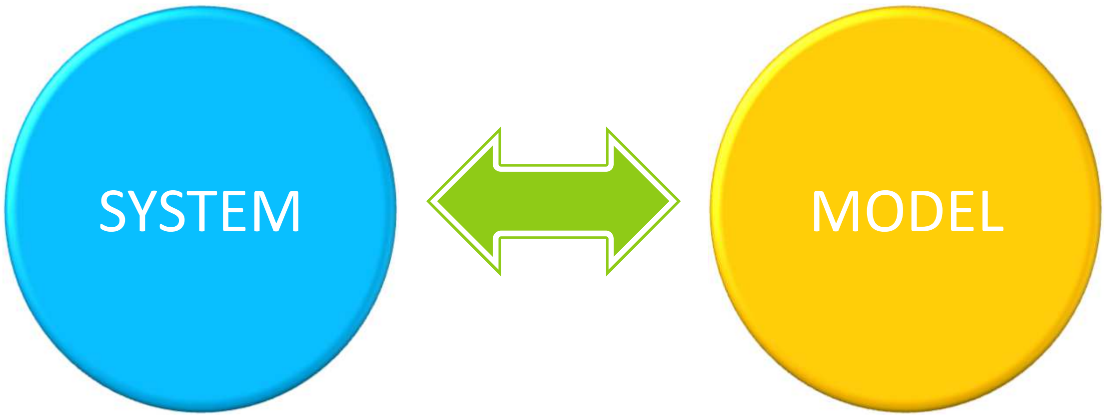


1.1 Systems, Models, and Modeling

- A **model** is *a description of a system*.
- A **system** is *any collection of interrelated objects*.
- An **object** is *some elemental unit upon which observations can be made, but whose internal structure either does not exist or is ignored*.
- A **description** is *a signal that can be decoded or interpreted by humans*.

1.1 Systems, Models, and Modeling

In short, systems are anything humans wish to discuss and models are one tool that facilitates the discussion.

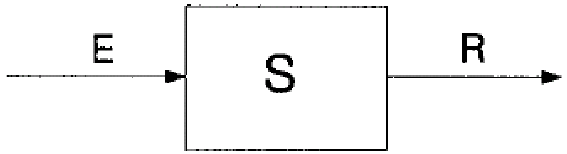


1.2 Uses of Scientific Models

- *Model [er]: a device for turning assumptions into conclusions.* - Schimel (2002)
- Three primary, technical uses of models in science:
  - **Understanding** – of either a real, physical system or of a system of logic such as another scientific theory.
  - **Prediction** – of the future or of some state that is currently unknown.
  - **Control** – to constrain or manipulate a system to produce a desirable condition.

1.2 Uses of Scientific Models

- Conceptual framework of systems that defines the three uses of models - Karplus (1983)



Type of Problem	Given	To Find	Uses of Models
Synthesis	E and R	S	Understand
Analysis	E and S	R	Predict
Instrumentation	S and R	E	Control

Figure 1.1: Systems and the uses of models. Top: A general system represented as an input (E), a system object (S), and the output (R). Bottom: Knowledge needed for models of different uses. (From Karplus 1977, Fig. 1. © 1977 Simulation Councils, Inc. Reprinted withOUT permission Simulation Councils, Inc., publisher.)

1.2 Uses of Scientific Models

- Three general problems that human face with respect to any discipline or body of knowledge are:
  - **Synthesis** – use knowledge of inputs and outputs to infer system characteristics.
  - **Analysis** - use knowledge of the parts and their stimuli to account for the observed responses.
  - **Instrumentation** - design a system such that a specified output is the result of an input.

1.2 Uses of Scientific Models

- Secondary uses of scientific models that derive from the social characteristics of science:
  - Use as a conceptual framework for organizing or coordinating empirical research (e.g., designing experiments or sampling studies, allocating limited research dollars).
  - Use as a mechanism to summarize or synthesize large quantities of data (e.g., a simple linear regression equation $y = mx + b$ to reduce all of the x-y pairs of data to two parameters m and b).
  - Identify areas of ignorance, especially when defining relations between objects (e.g., Does species A eat species B).
  - Provide "insight" to managers or planners (or others) by performing "what-if“ simulations ("gaming").

1.3 Example: Island Biogeography

- Physical Setting

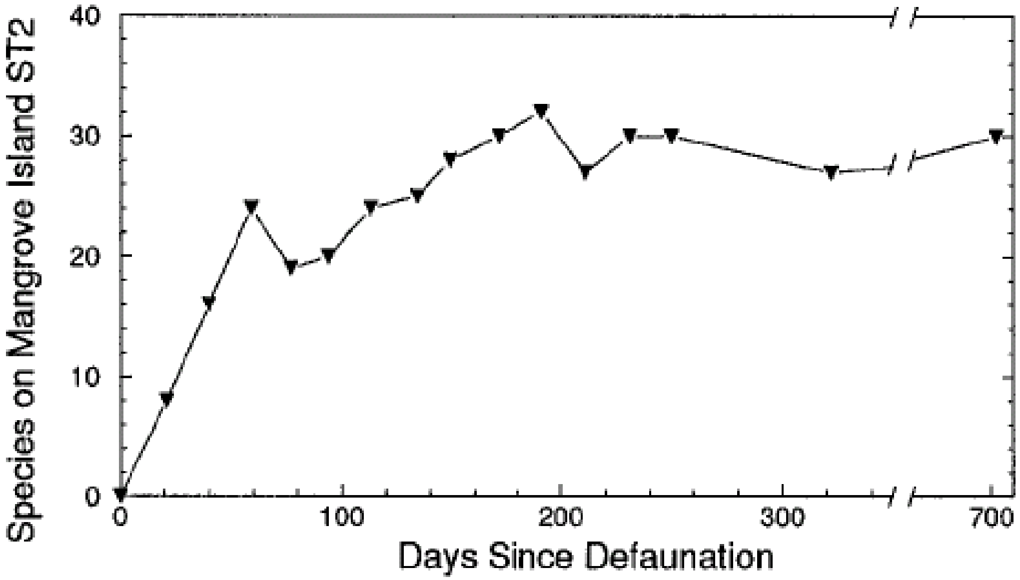


Figure 1.2: Numbers of insect species on a small mangrove island following defaunation. (From Simberloff and Wilson 1970, Fig. 1. © 1970 Ecological Society of America. Reprinted with permission of the publisher.)

1.3 Example: Island Biogeography

- Theory
  - The number of species is a balance of two processes: immigration and extinction.
  - The rate of both processes depend on the number of species currently on the island.
  - The net rate of change of species is the sum of these two "forces."

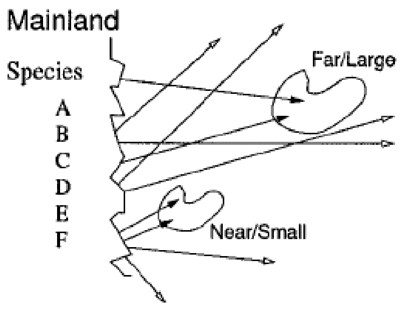


Figure 1.3: Physical picture of island biogeography theory. Organisms colonize randomly (arrows). Islands can vary by their distance to the mainland (near or far) and their size (large or small).

1.3 EXAMPLE: ISLAND BIOGEOGRAPHY

- Biological Hypotheses
  - Individuals of each species have a constant probability of arriving at the island and this probability is identical for all individuals and all species. The rate of immigration (I) of new species only occurs upon the arrival of an individual of a species not currently on the island.
  - The probability of extinction of any single species is constant. Consequently, as the number of species on the island increases, the probability that any one species goes extinct increases. Thus, the total rate of extinction (E) increases with R (number of species on the island).

1.3 Example: Island Biogeography

- Graphical Illustration of the Hypotheses

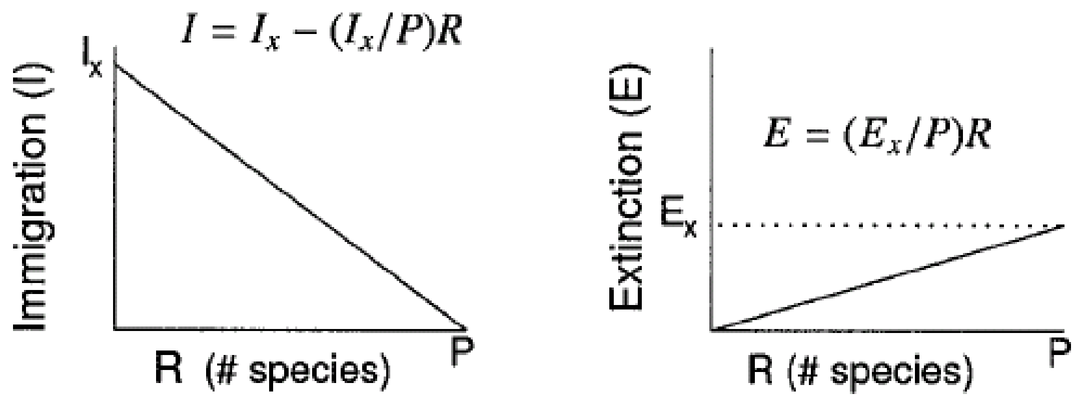


Figure 1.4: Quantitative relationships between number of species on an island (R) and rates of immigration (I) and extinction (E). P is the number of species in the mainland pool of species.

1.3 Example: Island Biogeography

- Mathematical Expression of the Hypotheses
  - Linear model

$$I = I_x - (I_x / P)R$$
$$E = (E_x / P)R$$

  - Assemble into one recursive finite difference equation

$$\begin{aligned} R_{t+1} &= R_t + I_t - E_t \\ &= R_t + I_x - (I_x / P)R_t - (E_x / P)R_t \qquad (1.1) \end{aligned}$$

1.3 EXAMPLE: ISLAND BIOGEOGRAPHY

- A Classroom Physical Simulation of the Colonization Process

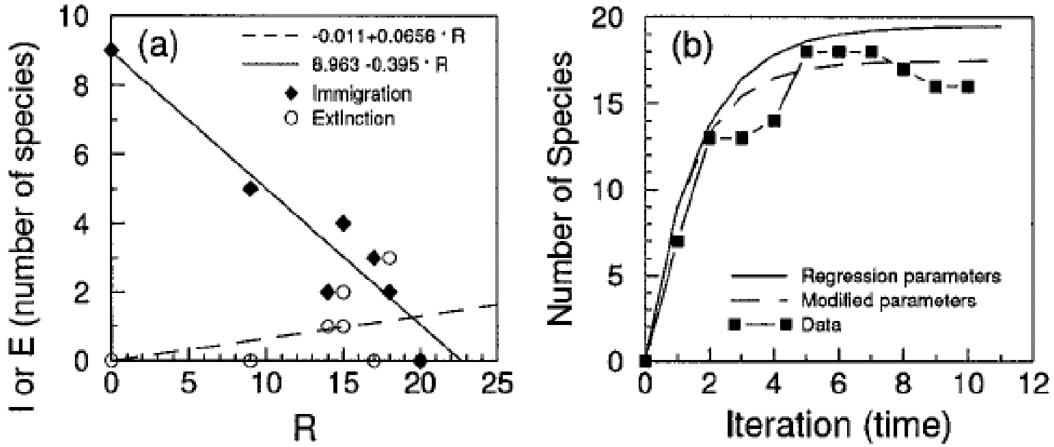


Figure 1.5: Data and results from a simulated biogeographical experiment. (a) Immigration rate (I, numbers/time, solid diamonds) and its best fit regression line (solid line). Also shown are Extinction rate (E, numbers/time, open circles) and its regression line (dashed line). (b) Observed and predicted number of species by iterating Eq. 1.2 using two estimates of parameters.

1.3 Example: Island Biogeography

- A Classroom Physical Simulation of the Colonization Process
  - Equation based on regression

$$R_{t+1} = R_t + (8.963 - 0.359R_t) - (-0.011 + (0.0656)R_t) \qquad (1.2)$$

  - Computing the equilibrium state

$$0 = I_x - (I_x / P)\hat{R} - (E_x / P)\hat{R}$$

  - Mechanism (Fig. 1.5a) vs. observable dynamics (Fig. 1.5b)

1.4 Classifications of Models

- Forms of Models
  1. **Conceptual or Verbal** - descriptions in a natural language.
  2. **Diagrammatic** - graphical representations of the objects and relations (e.g., ecological "box-and-arrow" diagrams of energy flow, physiological diagrams of metabolic pathways such as the Krebs cycle).
  3. **Physical** - a real, physical mock-up of a real system or object (either larger or smaller: a "tinker-toy" model of DNA or a scale model of an airplane for a wind tunnel).
  4. **Formal** - mathematical (usually using algebraic or differential equations).

1.4 Classifications of Models

Mathematical Classification

1. **Does the mathematics have an explicit representation of mechanistic processes?**

   YES: Process-oriented or mechanistic models (e.g., hydrology models using Newtonian physics and chemistry).

   NO: Descriptive or phenomenological models (e.g., the island biogeography model).

2. **Does the mathematics have an explicit representation of future system states or conditions?**

   YES: Dynamic models (e.g., island biogeography model).

   NO: Static models (e.g., linear regression equation relating variables x and y).

3. **Does the mathematics represent time continuously?**

   YES: Continuous models, time may take on any values (e.g., 3.3 sec).

   NO: Discrete models, time is an integer only.

1.4 Classifications of Models

Mathematical Classification

4. **Does the mathematics have an explicit representation of space?**
   - **YES: Spatially heterogeneous models (e.g., objects have a position in space, or occupy a finite region of space).**
     - **a) Discrete: space is represented as cells or blocks, and each cell is represented as spatially homogeneous.**
     - **b) Continuous: every point in space is different (e.g., diffusion equations in physics).**
   - **NO: Spatially homogeneous models (e.g., simple equations of population dynamics or enzyme kinetics).**
5. **Does the model allow random events?**
   - **YES: Stochastic models (e.g., random temperature values may produce random changes in the intrinsic rate of increase in population dynamics models.**
   - **NO: Deterministic models (i.e., all parameters constant).**

1.4 Classifications of Models

- Mathematical Classification
  - Example:

    The model of island biogeography (Eq. 1.1) is a *deterministic, spatially homogeneous, discrete time, descriptive, dynamic* model.

1.4 Classifications of Models

- System Concept Classification
  1. **Compartment model**- differential or finite difference equations.
  2. **Transport model** - partial differential equations.
  3. **Particle model** – fate of individual particles moving in space or individual organisms changing their condition.
  4. **Finite state automata** - represent an object as being in only a few, finite number of states or conditions.

1.5 Constraints on Model Structure

- ***Realism:*** the degree to which model structure mimics the real world. In formal models that are realistic, the equations are correct, not just the model output. In physical models (e.g., a scale airplane) maximal physical detail is present (i.e., every rivet).
- ***Precision:*** the accuracy of the model predictions (output). In precise models, the air flow around the scale model is exactly the same as that around the fullsize plane. Precision is not used here in the statistical sense, which refers to the degree of variability of a set of measurements.
- ***Generality:*** the number of systems and situations to which the model correctly applies. In physical models, a general scale airplane model applies to both a Piper Cub (small, single-engine aircraft) as well as a Boeing 747 (large, multiple-jet engine aircraft).

1.6 Some Terminology

Table 1.1: A few more terms.

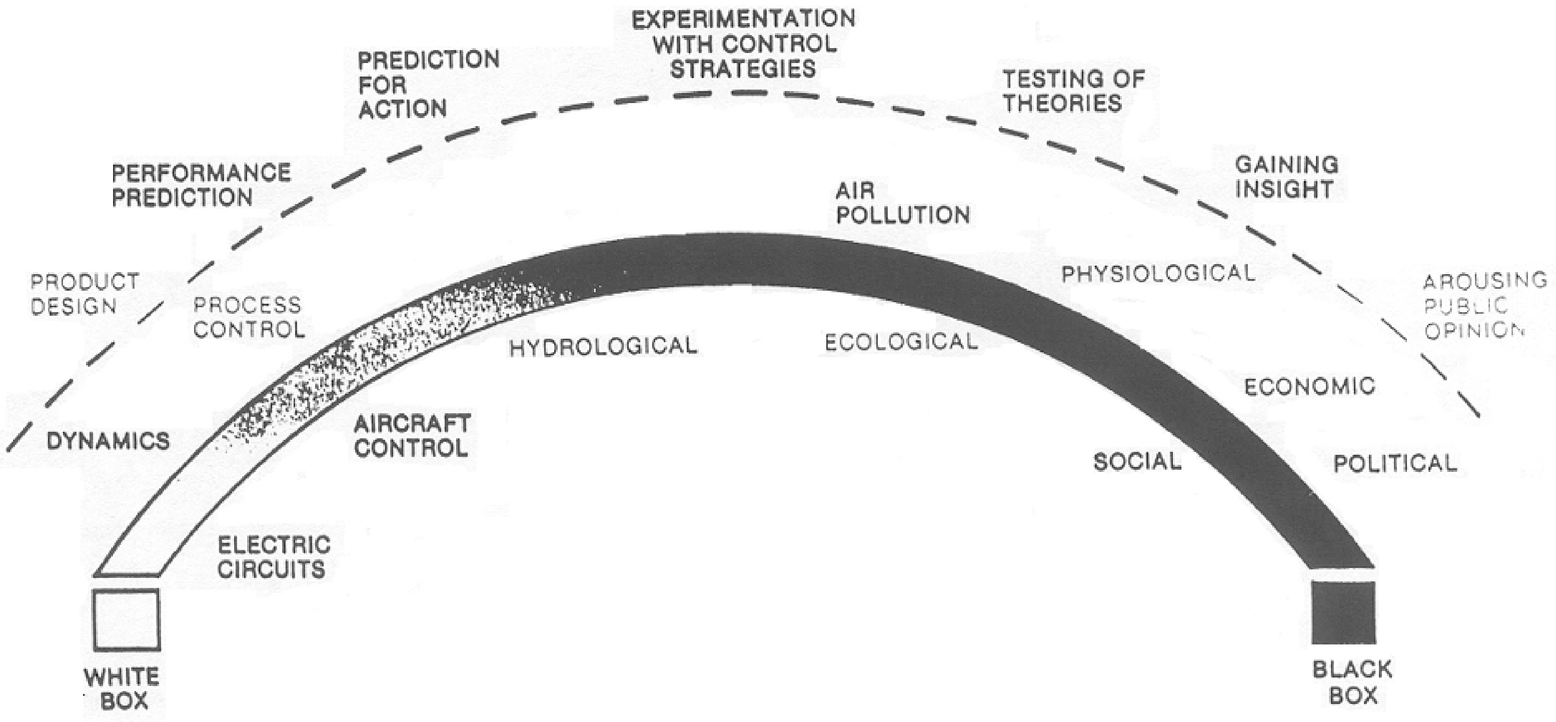
analytical model	*(n)* a mathematical model whose solution is not obtained by simulation or numerical approximation, but by purely mathematical argument or a model where mathematical properties (e.g., stability of equilibria) are achieved by mathematical argument
dynamic model	*(n)* a mathematical model that describes the changes over time of quantities representing the system objects (e.g., population sizes)
mathematical model	*(n)* a set of mathematical equations that describe a system
mathematical modeling	*(v)* the human activity of creating a set of mathematical equations that describe a system
model	(a) *(n)* a description of a system, (b) *(v)* the human activity of creating a description of a system
objective	*(n)* (a) the purpose for doing something, a goal, (b) a verbal statement that guides and constrains modeling, (c) a list including at least some of the following: objects and relations modeled, environment of the system modeled (influencing variables, objects not modeled), length of time that the model applies to the system, spatial and temporal scales of resolution, questions addressed of the model

1.6 Some Terminology

simulate	*(v)* (a) to produce a solution to a simulation model, (b) to model
simulation	*(n)* (a) a set of one or more numbers that together constitute a numerical solution to a simulation model, (b) one run of a computer program that numerically solves a simulation model
simulation model	*(n)* a mathematical model whose solution is obtained by numerical approximation, usually involving computers; not an analytical model
solution	*(n)* (a) an answer to a problem, (b) a set of numbers whose values satisfy a mathematical equation (e.g., the roots to a polynomial equation)
system	*(n)* a collection of objects and relations between objects
system state	*(n)* the set of particular, numerical values of all system objects at a given time (e.g., grams carbon in all species in an ecosystem)
well-defined system	*(n)* the smallest set of objects and relations whose states (values) cannot be proved to be unnecessary to achieve the objectives of the model

1.7 Misuses of Models: The Dark Side

- Motivations for modeling



(Karplus, 1983)

1.7 Misuses of Models: The Dark Side

- "White box" vs. "Black box"
- Quantitative model vs. qualitative model

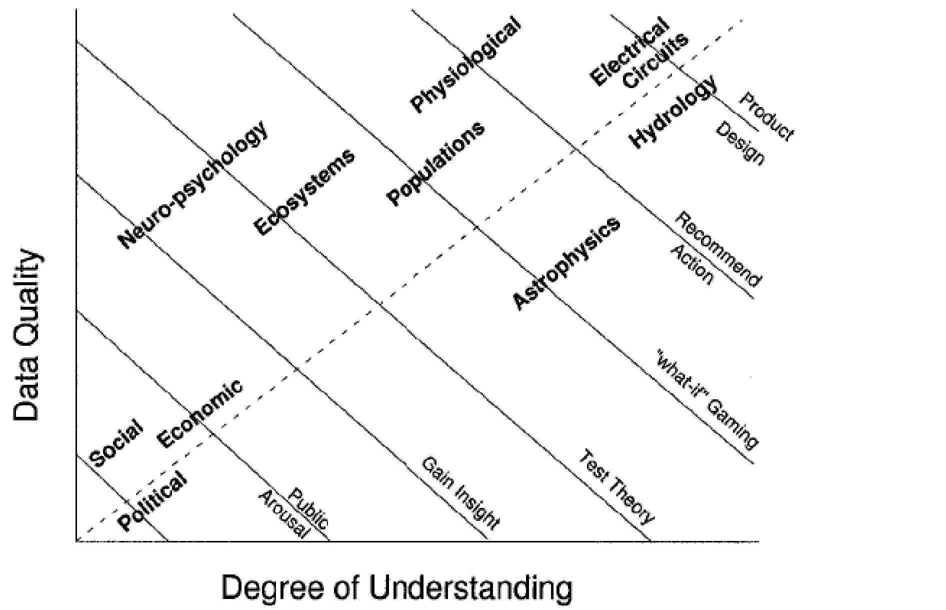


Figure 1.6: Appropriate uses of models related to degree of uncertainty in systems. Contour lines represent combinations of data quality and degree of understanding in disciplines for which models may be used as indicated on the lines. The dotted line is a continuum of uncertainty (lower left: much, upper right: little).